

Usman Asad

Senior AI Integration Engineer · Co-Founder & Technical Lead @ DigifyR

Islamabad, Pakistan · +92 313 5149287 · usmanasad0324@gmail.com · usman.digifyr.com · linkedin.com/in/usmanasad87 · github.com/UsmanAsad87 · Upwork: **Top Rated · 100% Job Success**

SUMMARY

Senior AI Integration Engineer who puts AI into real products — and finishes what AI tools started. My AI work is **live in production today**: an AI data-loss-prevention platform running a **self-hosted LLM** on GPU (ZeusLock.ai), **three AI agents** operating a HIPAA healthcare platform (a scheduling agent that cut overtime 23%), and a marketplace where a **vision model diagnoses car problems from a photo**. 40+ shipped products across security, healthcare, fintech and logistics as co-founder of DigifyR. Deep stack: **Python / FastAPI, .NET, Flutter, Next.js, PostgreSQL, LLM integration (llama.cpp, LangChain / LangGraph, Anthropic & OpenAI SDKs), Kubernetes, GitOps CI/CD** — from model to mobile app to cluster, one engineer.

PROFESSIONAL EXPERIENCE

Lead Developer | ZeusLock.ai

Nov 2025 – Present · Remote / France

AI-native Data Loss Prevention platform stopping sensitive-data leaks into AI tools (ChatGPT, Claude, Copilot, MCP); SaaS + air-gapped on-premise for European enterprise and sovereign clients — zeuslock.ai

- **Built the AI detection engine**: hybrid pipeline of regex rules, policy-aware LLM classification and a jailbreak classifier analyzing prompts and files (PDF, DOCX, XLSX, images) in real time.
- **Self-host the LLM stack** (Gemma via llama.cpp on NVIDIA GPU) behind an OpenAI-compatible API — fully sovereign EU inference; added multimodal detection and an OCR-first pipeline measured **3.3x faster** than vision-model inference.
- Cut LLM latency **~1.9x** with a hybrid reasoning policy, gated by a **602-case detection benchmark** that validates every model and prompt change — the discipline that keeps AI features from silently regressing.
- Own the multi-tenant FastAPI backend: PostgreSQL 16 with **Row-Level Security**, Celery, Keycloak (OIDC), encrypted API keys, audit logging, format-preserving anonymization, SIEM integrations (Splunk, Sentinel, Elastic/OCSF).
- Run the infrastructure: Kubernetes (k3s) on GPU, universal Helm chart, **ArgoCD GitOps**, CI/CD with image signing and SBOM; ship SaaS and air-gapped on-prem (Ed25519 licensing) from one codebase.
- Also deliver the Next.js dashboards, Manifest v3 browser extension and Electron desktop agent (MITM proxy, MCP monitoring, Shadow-AI discovery).

Co-Founder & Technical Lead | DigifyR

Jul 2022 – Present · Islamabad

Four-person AI product studio — 40+ products shipped for clients in healthcare, fintech, logistics and e-commerce; Top Rated on Upwork with 100% Job Success across all engagements — digifyr.com

- **HomeHealth.ai** — AI-native healthcare SaaS (HIPAA, AWS) with **three AI agents live in production**: certification-aware scheduling (cut overtime **23%** at full coverage), HR assistant, and compliance document intelligence. 31 organizations, 913 clinicians.
- **Mechanic Connect** — three-sided auto-service marketplace: 3 Flutter apps on one .NET API, **6 live store listings**, with **AI photo triage**: a vision LLM reads the photo, suggests the diagnosis and routes the request.
- **Mobileriz** — marketplace-ops platform for Turkish e-commerce (Trendyol, Hepsiburada): FastAPI + PostgreSQL + Redis, e-invoice SOAP integration, **AI product-listing generation on the Anthropic SDK**; a 568-hour engagement.
- **Ataya** (leisure marketplace, 10K+ downloads, +40% bookings) · **NafaPay** (mobile-money BNPL fintech, .NET 9 hexagonal, 1,400+ users) · **CargoControl** (maritime discharge control; largest tracked shipment 48,427 t) · **Maman Tontine** (digitised savings circles with Wave/Orange Money payouts).
- Set technical direction across the studio — architecture, code review, release cadence — and take AI-built apps (Lovable, Bolt, Cursor, FlutterFlow) from “80% done” to launched: real auth, real backends, through App Store and Play review.

Senior Python Developer | Aim Digital

Jun 2025 – Oct 2025 · Islamabad

Machine Monitoring System — mms-aim.netlify.app

- FastAPI + PostgreSQL microservices with real-time telemetry over WebSockets (Redis, RabbitMQ); predictive-maintenance alerts (WhatsApp/SMS/email) via Celery; Keycloak RBAC for 1,000+ users.

- Optimized API performance **20s** → **4s**; Docker + Azure App Service/ACR CI/CD; analytics dashboards cut manual reporting 40%.

Assistant Manager – Software Engineer | NESCOM

Jul 2023 – May 2025 · Islamabad

- Intelligent mission-planning system with genetic-algorithm path planning (**33% efficiency gain**); ArcGIS geospatial visualization; high-performance C++ TERCOM navigation utility for Windows/Linux.

SELECTED PROJECTS & RESEARCH

- FaceRT** — real-time multi-face recognition at **98.6% accuracy** (RetinaFace + ArcFace, TensorFlow; Flask API, MongoDB, React + Flutter clients). github.com/UsmanAsad87/FaceRecogFYPMModel
- Deep-learning stock-price prediction (published, 2023)** — proposed Single LSTM Engine, RMSE < 3% on S&P 500.

TECHNICAL SKILLS

AI Engineering: LLM integration (Anthropic & OpenAI SDKs, llama.cpp, OpenAI-compatible self-hosting), AI agents, RAG, LangChain, LangGraph, fine-tuning (Unsloth), prompt & guardrail engineering, jailbreak detection, vision + OCR pipelines, evaluation benchmarks & cost controls, TensorFlow/Keras, scikit-learn

Backend: Python (FastAPI, SQLAlchemy, Alembic, Pydantic, Celery, asyncio), .NET / C#, Node.js, microservices, Clean/Hexagonal Architecture, WebSockets, RabbitMQ, Redis, Keycloak (OIDC/JWT)

Mobile & Web: Flutter (40+ store releases), Next.js, React, TypeScript, Tailwind

Data: PostgreSQL (RLS, multi-tenancy), MySQL, MSSQL, MongoDB, Firebase

Cloud & DevOps: Kubernetes (k3s), Helm, ArgoCD GitOps, Docker, GitHub Actions (signing, SBOM), AWS, Azure, GCP, OVH, Traefik/nginx

Security & Payments: OIDC/OAuth2, RBAC, SIEM (Splunk, Sentinel, Elastic), GDPR/HIPAA-aware design, Stripe, Wave, Orange Money

EDUCATION & AWARDS

BSc Computer Science — PIEAS, Islamabad · 2019–2023 · GPA **3.91/4.00** · Fully-funded NESCOM Fellowship
Hafeez Qureshi NESCOM Excellence Award (2024) · Co-Lead ML, Google Developer Student Club · Treasurer, PIEAS AI Society